

Ihsan ULLAH

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🏠 Date of Birth: May 05, 1995 — 📍 Place of Birth: Lower Dir, Pakistan



Interests —

Artificial Intelligence, Data Science, Datasets, Competitions/Benchmarks, Software Engineering

Skills

Languages Python, Swift, Javascript, PHP

Machine Learning Data processing and analysis, Datasets preparation, Challenge Organization

ML Tools Pytorch, Tensorflow, Keras, Scikit-Learn, Pandas, Numpy

Cloud GCP(Compute Engine, Cloud Storage)

Web HTML, CSS, Bootstrap, JavaScript, React, Django

Databases MySQL, PostgreSQL, NoSQL, Firebase

Technical Writing Research/Project proposals, Technical reports, Conference papers

Other Docker, CircleCI, Git/GitHub, OpenCV, ~~W~~TeX

Experience

ChLearn U.S.A

Nov 2022 – Present

Research Software Engineer

- **Data:** Data analysis, preprocessing, formatting, visualization, and datasets curation.
- **Competitions and Benchmarks:** Competitions/benchmarks design, metrics, leaderboards, tasks, evaluation, GitHub repository management, setting up competitions/benchmarks on open-sourced platforms (CodaLab and Codabench), and setting up compute workers on Google Cloud Platform. Creating competition bundles, starting kit Jupyter notebooks, and Google Colab notebooks. Creating Docker images for specific competition/benchmark needs.
- **Project Multitasking:** Managing concurrent and cross-disciplinary projects, managing deliverables, and meeting deadlines. Developing project websites for project visibility.
- **Solution Development:** Designing solutions using popular machine learning frameworks such as PyTorch, TensorFlow, and Scikit-learn. Writing Python code tailored to project needs, ensuring scalability and efficiency.
- **Open-source development:** Contributing as a core developer to the open-sourced Codabench platform to maintain it by fixing reported issues, adding new features, and improving functionality based on user feedback.
- **Writing and Documentation:** Writing proposals, reports, and research papers. Preparing documentation for projects, competitions, and benchmarks to ensure clarity and accessibility.
- **LLMs for Research:** Utilizing Large Language Models (LLMs) for research tasks such as in peer-review process for helping authors and reviewers, using prompt engineering, and automating processes for better efficiency and scalability.

Crédit Agricole SA, Paris France

May 2022 – Oct 2022

Data Science Intern (Data Analytics Team)

- Fairness in Machine Learning and AutoML
- Bias detection and mitigation

LISN, Université Paris-Saclay, France

May 2021 – Aug 2021

Artificial Intelligence Intern

- Datasets preparation and ML Challenge Organization
- Transfer Learning, Meta-Learning, Computer Vision

Université Paris-Saclay, France

Jan 2021 – Feb 2021

Travail d'étude et de recherche

- Data preprocessing
- Challenge Design and Organization

Codematics Inc, Abbottabad Pakistan

Jan 2020 – Aug 2020

Software Project Manager

Codematics Inc, Abbottabad Pakistan

Jan 2019 – Dec 2019

Software Engineer

Ocheng/Chinaccelerator, Shanghai China

May 2018 – Dec 2018

Education

Master's in Artificial Intelligence

Université Paris-Saclay, France

Scholarship: Labex Digicosme Excellence Scholarship

Sep 2020 – Nov 2022

Bachelor of Science in Software Engineering

University of Engineering and Technology Peshawar, Pakistan

CGPA: 3.70/4.0

Award: University Gold Medal – Graduated with Distinction

Sep 2013 – Aug 2017

Intermediate (High School)

Govenment Post-Graduate College Mardan, Pakistan

Marks: 839/1100

Sep 2011 – Aug 2013

Publications

Stylized Meta-Album: Muti-domain computer vision meta-dataset

Journal of Data-centric Machine Learning Research – 2024

<https://stylized-meta-album.github.io/>

Submitted

Usefulness of LLMs as an Author Checklist Assistant for Scientific Papers: NeurIPS'24 Experiment

<https://arxiv.org/pdf/2411.03417>

FAIR Universe HiggsML Uncertainty Challenge Competition

NeurIPS 2024 Competition Track – 2024

<https://arxiv.org/pdf/2410.02867>

RelevAI-Reviewer: A Benchmark on AI Reviewers for Survey Paper Relevance

Conf 'erence sur l'Apprentissage automatique (CAp) – 2024

<https://hal.science/LISN/hal-04608255v1>

Meta-Album: Multi-domain Meta-Dataset for Few-Shot Image Classification

Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks – 2022

<https://meta-album.github.io/>

Cross-Domain MetaDL: Any-Way Any-Shot Learning Competition with Novel Datasets from Practical Domains

NeurIPS 2022 Competition Track – 2022

<https://metalearning.chalearn.org/#h.1u3w2mh9pbjf>

Lessons learned from the NeurIPS 2021 MetaDL challenge

PMLR - Proceedings of the NeurIPS 2021 Competition and Demonstration Track – 2022

<https://proceedings.mlr.press/v176/el-baz22a.html>

MetaDL: Few Shot Learning Competition with Novel Datasets from Practical Domains

NeurIPS 2021 Competition Track – 2021

<https://metalearning.chalearn.org/metadlneurips2021>

Projects

Codabench

Open-sourced platform fo Competitions and Benchmarks

<https://www.codabench.org/>

<https://github.com/codalab/codabench>

- Codabench is an open-sourced platform organizing competitions and benchmarks related to Machine Learning and Data Science. This platform is a complete web based system with front-end developed in React.js and back-end developed in Django. It allows users to connect their own compute power for processing submissions submitted to competitions/benchmarks. This platform and project is lead by CodaLab at LISN and Université Paris-Saclay.

NeurIPS Checklist Assistant

An LLMs based checklist assistant for NeurIPS submissions

<https://www.codabench.org/competitions/2338/>

<https://blog.neurips.cc/2024/05/07/soliciting-participants-for-the-neurips-2024-checklist-assistant-study/>

- This study explores the use of Large Language Models (LLMs) as an assistant to help authors verify their submissions against the NeurIPS Paper Checklist. The goal is to assess whether LLMs can improve submission quality at NeurIPS. Participants receive feedback from an experimental LLM assistant to check compliance with NeurIPS submission standards. The LLM provides detailed feedback on the checklist responses to help authors refine their papers before submission. While the tool offers valuable guidance, it is meant to complement, not replace, the author’s judgment and expertise.

FAIR Universe

Unbiased Data Benchmark Ecosystem for Physics

<https://fair-universe.lbl.gov/>

- The FAIR Universe project, funded by the US Department of Energy, is a collaboration between Lawrence Berkeley National Laboratory, Université Paris-Saclay, University of Washington, and ChaLearn. The initiative aims to create a large-scale AI platform for hosting scientific datasets, models, and machine learning competitions to advance discoveries in high energy physics and cosmology. The project focuses on reducing systematic uncertainties in High Energy Physics through a series of challenges. Key events include a toy challenge (October 2023), a Particle Physics hackathon (November 2023), and the HiggsML Uncertainty Pilot Competition (March 2024). A major challenge on uncertainties in fundamental science is set to launch at NeurIPS 2024. The project is ongoing and will conclude in 2025.

Stylized Meta-Album

Muti-domain computer vision meta-dataset

<https://stylized-meta-album.github.io/>

- The Stylized Meta-Album (SMA) is a new image classification meta-dataset featuring 24 datasets (12 content and 12 stylized) to support research in out-of-distribution (OOD) generalization and related areas. SMA combines diverse subjects and styles, creating 4800 groups that offer extensive variability for rigorous studies. It introduces benchmarks for OOD generalization and group fairness, as well as unsupervised domain adaptation (UDA), showing the importance of group diversity in fairness and algorithm rankings, while also reducing error bars in benchmarking scenarios.

Meta-Album

A meta-dataset for few-shot image classification

<https://meta-album.github.io/>

- The Meta-Album is an image classification meta-dataset designed to support tasks such as few-shot learning, transfer learning, and meta-learning. It consists of 40 diverse open datasets from domains like ecology, manufacturing, human actions, and optical character recognition, each containing at least 20 classes with 40 examples per class. The datasets are uniformly preprocessed and available in three versions (Micro, Mini, Extended) to accommodate different computational needs. Meta-Album, already larger and more diverse than similar efforts, continues to expand through a series of competitions, creating a rolling benchmark for research.

References

Isabelle Guyon

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President, ChaLearn

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Lisheng Sun Hosoya

Postdoctoral Researcher, LISN Université Paris-Saclay

Researcher, ChaLearn

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Engineer, ASARD (LISN)

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Adrien Pavao

Founder, MLChallenge

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